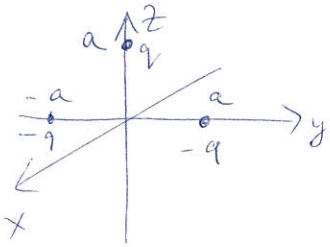
P3.32



for $r \gg a$, find $\vec{E}(r)$.

Multipole expansion (equs. 3.96 and 3.98, 3.99)

$$V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\left\{\frac{1}{r}Q + \frac{\vec{p}\cdot\hat{r}}{r^2} + \dots\right\}$$

$$Q = \int \rho(r')\,d^3r' = \text{total charge}$$

$$\vec{p} = \int \vec{r}\,'\,\rho(\vec{r}\,')\,d^3r' = \text{dipole moment}$$

Here $\rho(r') = -q\,\delta(x)\delta(z)\left(\delta(y-a)+\delta(y+a)\right) + q\,\delta(x)\delta(y)\delta(z-a)$

$$\Rightarrow Q = \int \rho(r')\,dx\,dy\,dz = -2q + q = -q$$

$$\vec{p} = \int \left(x\hat{x} + y\hat{y} + z\hat{z}\right)\rho(r)\,dx\,dy\,dz = (-q)\left(a+(-a)\right)\hat{y} + qa\hat{z}$$
$$= qa\hat{z}.$$

Spherical coordinates $\hat{z} = \cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}$

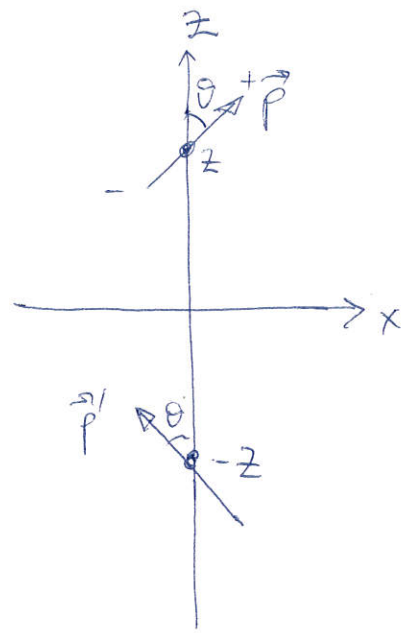
$$\Rightarrow \vec{p}\cdot\hat{r} = qa\cos\theta$$

$$\Rightarrow V(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\left(\frac{-q}{r} + \frac{qa\cos\theta}{r^2}\right)$$

$$\vec{E}(\vec{r}) = -\vec{\nabla}V = -\left(\hat{r}\frac{\partial}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial}{\partial\theta} + \hat{\varphi}\frac{1}{r\sin\theta}\frac{\partial}{\partial\varphi}\right)V(\vec{r})$$

$$= \frac{1}{4\pi\varepsilon_0}\left\{\left(\frac{-q}{r^2} - \frac{2qa\cos\theta}{r^3}\right)\hat{r} + \frac{qa\sin\theta}{r^3}\hat{\theta}\right\}$$

P4.6



Dipole $\vec{p} = p(\cos\theta\,\hat{z} + \sin\theta\,\hat{x})$

Image dipole $\vec{p}' = p(\cos\theta\,\hat{z} - \sin\theta\,\hat{x})$

Electric field of the (eq. 3.104) image dipole at point z:

$$\vec{E}' = \frac{1}{4\pi\varepsilon_0}\frac{1}{(2z)^3}\left(3(\vec{p}'\cdot\hat{z})\hat{z} - \vec{p}'\right)$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{1}{(2z)^3}\left(2p\cos\theta\,\hat{z} + p\sin\theta\,\hat{x}\right)$$

Torque created by $\vec{E}'$ is $\vec{N} = \vec{p}\times\vec{E}'$ (eq. 4.4)

$$\vec{N} = \frac{1}{4\pi\varepsilon_0}\frac{1}{(2z)^3}\left(p\cos\theta\,\hat{z} + p\sin\theta\,\hat{x}\right)\times\left(2p\cos\theta\,\hat{z} + p\sin\theta\,\hat{x}\right)$$

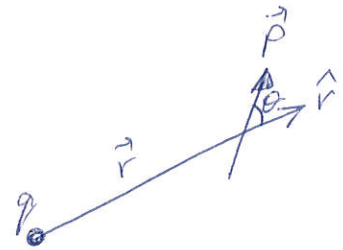
$$\hat{z}\times\hat{x} = \hat{y}\ ,\ \hat{x}\times\hat{z} = -\hat{y}$$

$$\vec{N} = \frac{1}{4\pi\varepsilon_0}\frac{p^2}{(2z)^3}\underbrace{\left(\cos\theta\sin\theta\,\hat{y} - 2\sin\theta\cos\theta\,\hat{y}\right)}_{-\sin\theta\cos\theta\,\hat{y} = -\frac{1}{2}\sin(2\theta)\,\hat{y}}$$

$$\vec{N} = \frac{1}{4\pi\varepsilon_0}\frac{p^2\sin(2\theta)}{2(2z)^3}\left(-\hat{y}\right)$$

Note that $-\hat{y}$ is "out-of-page" direction.

P4.9



(a) $\vec{F}_a = (\vec{p}\cdot\vec{\nabla})\,\vec{E}$

↳ el. field of charge q at point $\vec{r}$

$$\vec{F}_a = (p_x\hat{x} + p_y\hat{y} + p_z\hat{z})\cdot\left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)\left(\frac{q\,\vec{r}}{4\pi\varepsilon_0 r^3}\right)$$

$$= \left(p_x\frac{\partial}{\partial x} + p_y\frac{\partial}{\partial y} + p_z\frac{\partial}{\partial z}\right)\frac{q}{4\pi\varepsilon_0}\,\frac{x\hat{x}+y\hat{y}+z\hat{z}}{(x^2+y^2+z^2)^{3/2}}$$

$$= \frac{q}{4\pi\varepsilon_0}\left\{p_x\left[\frac{\hat{x}}{(x^2+y^2+z^2)^{3/2}} - \frac{3}{2}(2x)\,\frac{x\hat{x}+y\hat{y}+z\hat{z}}{(x^2+y^2+z^2)^{5/2}}\right]\right.$$

$$+ p_y\left[\frac{\hat{y}}{(x^2+y^2+z^2)^{3/2}} - \frac{3}{2}(2y)\,\frac{x\hat{x}+y\hat{y}+z\hat{z}}{(x^2+y^2+z^2)^{5/2}}\right]$$

$$\left. + p_z\left[\frac{\hat{z}}{(x^2+y^2+z^2)^{3/2}} - \frac{3}{2}(2z)\,\frac{x\hat{x}+y\hat{y}+z\hat{z}}{(x^2+y^2+z^2)^{5/2}}\right]\right\} =$$

$$= \frac{q}{4\pi\varepsilon_0}\left\{\frac{p_x\hat{x}+p_y\hat{y}+p_z\hat{z}}{(x^2+y^2+z^2)^{3/2}} - \frac{3\,(p_x x + p_y y + p_z z)\,(x\hat{x}+y\hat{y}+z\hat{z})}{(x^2+y^2+z^2)^{5/2}}\right\}$$

$$= \frac{q}{4\pi\varepsilon_0}\left\{\frac{\vec{p}}{r^3} - \frac{3(\vec{p}\cdot\vec{r})\,\vec{r}}{r^5}\right\} = \frac{q}{4\pi\varepsilon_0}\,\frac{\vec{p} - 3(\vec{p}\cdot\hat{r})\,\hat{r}}{r^3}$$

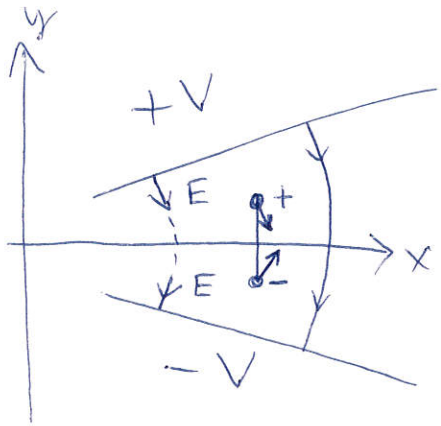
Clearly $\vec{F}_a = -\vec{F}_b$ as it must be by Newton's 3rd law.

(b) force from dipole on the charge

$$\vec{F}_b = q\,\vec{E}_{dip} = q\,\frac{1}{4\pi\varepsilon_0}\,\frac{3(\vec{p}\cdot(-\hat{r}))(-\hat{r}) - \vec{p}}{r^3} = \frac{q}{4\pi\varepsilon_0}\,\frac{3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}}{r^3}$$

↳ el. field of the dipole, unit vector is now in $-\hat{r}$ direction

P4.30



El. field must be perpendicular to equipotential surfaces

$\Downarrow$

dipole experiences <u>net</u> force to the right, along $+\hat{x}$ axis.